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RESEARCH ARTICLE

Machine Learning-Based Enhanced Embryo Selection Using Temporal Feature Extraction and Rule-Based Labeling Model With Decision Trees

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ABSTRACT Automated embryo quality prediction remains a critical challenge in Assisted Reproductive Technology (ART) due to the subjective nature of manual grading and the lack of methods that integrate temporal developmental patterns. Embryo imaging datasets are scarce and typically limited in size. Existing approaches often neglect the time-series behavior of embryos, discard unlabeled embryos, and overlook the predictive utility of unlabeled data. Deep learning models require massive amounts of data to learn features. To overcome these limitations, we introduce an integrated framework that combines NASNet-based stage annotation, forward-filled time-aligned features, decision tree-based label imputation, and a fine-tuned SMOTE-augmented XGBoost classifier. To address the issue of missing labels, we introduce a novel rule-based decision-tree imputation strategy. It leverages domain-driven criteria and temporal features to infer missing labels, thereby enhancing dataset completeness for deep learning models. This integrated approach captures temporal developmental patterns and morphological quality while addressing class imbalance and label sparsity. Our model achieves superior predictive performance, with an accuracy of 90.1%, a macro F1-score of 0.844, and a weighted F1-score of 0.902, outperforming the baseline XGBoost and standard SMOTE variants. Specifically, without feature engineering or appropriate handling of missing values, the macro F1-score was only 0.479; with feature engineering alone without imputation, it is reduced to 0.356; however, with feature engineering combined with decision tree imputation, performance improved to a macro F1 of 0.844. Additionally, SHAP-based interpretability analysis reveals biologically meaningful features contributing to predictions. These results demonstrate the potential of our framework to support objective, interpretable, and data-efficient embryo selection in ART workflows.

INDEX TERMS Embryo quality prediction, time-series analysis, label imputation, decision tree, NASNet, SMOTE, XGBoost.

I. INTRODUCTION

Since the first successful in vitro fertilization (IVF) in 1978, over 8 million babies have been born globally through assisted reproductive technologies (ART), with current global live birth rates per IVF cycle ranging from 30% to 35% [1]. However, the challenge of accurately selecting viable embryos for transfer persists, as suboptimal selection contributes to lower success rates and multiple cycles, increasing patient costs and emotional stress [1]. Research into automated, data-driven embryo assessment is crucial for improving pregnancy outcomes and reducing the burden on patients, making it a priority in ART advancements [1].

Recent studies have leveraged deep learning models, such as convolutional neural networks (CNNs), for embryo quality assessment, with approaches including transfer learning for blastocyst classification and hyperparameter optimization for

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small datasets [2], [5]. For example, Khosravi et al. applied deep learning to assess blastocyst quality, Chai and Gunawan optimized CNNs for small datasets, and Thirumalaraju et al. used CNNs to classify embryo morphology [2], [3], [5]. However, these models often rely on static images at specific time points, overlooking the temporal dynamics of embryo development, and frequently discard data with missing labels, reducing dataset robustness [2]. Additionally, their lack of interpretability limits clinical adoption due to the opaque nature of deep learning predictions [1].

The objective of this research is to develop a comprehensive pipeline for predicting blastocyst quality in ART by integrating time-lapse image analysis with advanced machine learning techniques. The scope includes preprocessing embryo images, annotating developmental stages, extracting temporal features across standardized time intervals, handling missing labels, and classifying embryos as ''Good'' or ''Not Good'' for blastocyst potential [4]. This approach aims to capture the dynamic progression of embryo development while ensuring interpretability for clinical applicability.

While there have been advances in deep learning for assessing embryo observations, unresolved issues remain. The first issue is that most existing models rely solely on static images and therefore fail to incorporate the temporal dynamics of early embryo development; those changes are critical for clinicians to assess. Another key issue is that, in practice, available datasets often show that many embryos are 'discarded from analysis' during training, which decreases generalizability to clinical practice. Most high-performing deep learning models have limited interpretability, offering little insight into their decision processes, which poses challenges for embryologists who must justify embryo grading decisions. As dataset size increases, classifiers may exhibit bias toward overrepresented high-quality embryos. The above issues regarding core principles, such as the incorporation of time as a variable, the completeness of labels, the need for transparency, and the unevenness in modeling, remain unaddressed.

The motivation for this research stems from the critical need to enhance embryo selection accuracy in ART, as current manual assessments by embryologists are subjective and time-consuming, leading to variability in outcomes [1]. The underutilization of time-lapse imaging data's temporal information and the exclusion of embryos with incomplete data in existing studies limit the potential for robust, scalable solutions, driving the need for innovative approaches to improve IVF success rates [2], [5].

The main contribution of the research is:

- Temporal feature extraction: extracts stage-specific ratios and grade distributions across standardized 30-minute intervals 24–64 hpi), capturing dynamic developmental patterns.
- Rule-based label imputation: uses a decision tree to impute missing labels, retaining all embryos for analysis, unlike studies that discard incomplete data.
- Fine-tuned SMOTE: implements a tailored Synthetic Minority Over-sampling Technique (SMOTE) strategy ($sampling_strategy = 0.8$, k_neighbors $= 3$) to address class imbalance, improving minority class representation.
- Hybrid model integration: combines NASNet for stage annotation with XGBoost for quality prediction, leveraging complementary strengths.

II. LITERATURE SURVEY

Comparative deep learning model [6] utilizing VGG 19, ResNet, MobileNet, and EfficientNet B0 to evaluate inner cell mass and trophectoderm quality from embryo images. Harnessing data augmentation and class-balanced training, the study thoroughly trained on 1,020 images from different developmental stages to overcome small sample sizes and class imbalance. Grad-CAM was deployed to demonstrate model attention, thereby helping validate biological relevance. EfficientNet B0 emerged as the most accurate and robust over tests. Limitations such as the relatively small, homogeneous image set and dependence on static captures potentially neglect temporal morphological cues. The study also highlighted the need for external clinical validation and the integration of dynamic developmental information. Automated 3D convolutional neural network [7] trained on time-lapse sequences from days 2, 3, and $\geq$5, using an expansive dataset of over 181,000 embryos across 22 clinics. The model accomplished AUCs between 0.62–0.71 when predicting implantation results and notably surpassed manual grading tools at the blastocyst stage. It also integrated calibration to correlate model scores with actual implantation probabilities, thereby guiding clinical interpretation. However, performance was lower at early cleavage stages, underscoring a temporal limitation. Additionally, the 3D CNN remains largely non-interpretable, which limits interpretability, and its reliance on time-lapse hardware may restrict adoption in standard clinical settings.

Convolutional neural networks in the article [8] were implemented for time-lapse embryo imaging, mining accuracy, sensitivity, and AUC over 22 retrospective studies analyzing across 522,000 images from roughly 223,000 embryos. The review emphasized robust CNN performance, with most models surpassing 80% accuracy and even outperforming embryologists in some instances. Methodologically, studies utilized diverse CNN architectures and handled morphokinetic patterns captured at frequent intervals. However, heterogeneity was greatest due to varied datasets, labelling criteria, and clinical biases; 10 studies indicated an elevated risk of patient selection bias. Additionally, standardization issues and reporting inconsistencies restrict reproducibility and comparability. The review recommended greater uniformity in data sharing and harmonization of protocols to boost clinical translation. Hybrid method [9] integrating discrete wavelet transform (DWT) for feature

extraction with transfer learning from pre-trained CNN backbones to classify human blastocyst quality from static images. Wavelets improved spatial-frequency discrimination, and transfer learning fine-tuning addressed the problem of limited dataset size. While optimizing classification accuracy, the static-image method overlooked dynamic cues from embryo development. The study was limited by a modest sample size and limited external validation, increasing concerns about generalizability over imaging systems and lab protocols.

CNN [10] trained on static images of mature oocytes to predict subsequent blastocyst development, enabling viability forecasting before fertilization. By exploiting oocyte morphological features, the model achieved moderate predictive accuracy but was hindered by limited interpretability and a small sample size. It did not integrate dynamic morphokinetic data and was developed using images from only two clinics, limiting its broader clinical relevance and requiring validation across diverse populations. Deep convolutional neural network with boosting methods [11] for predicting quality in bovine embryos, with a focus on animal reproduction contexts. Boosting increased ensemble performance by iteratively highlighting misclassified instances. While this technique boosted accuracy, animal-specific data, small sample size, and different imaging limitations hinder its applicability in human IVF. The model also did not address temporal development or generalize over labs.

Comparative analysis [12] evaluating EfficientNet B0 alongside other CNNs (ResNet, Inception, VGG) for five-class embryo grading by static images. EfficientNet B0 achieved top-tier accuracy while minimizing computational cost, demonstrating parameter efficiency in multi-class grading tasks. A key limitation is the reliance on static images, which prevents the utilization of time-lapse developmental dynamics, and a single-centre dataset without external validation, potentially influencing generalizability. Morphology euploidy correlation [13], correlating embryo morphological grading, clinical patient parameters, and euploidy outcomes assessed through preimplantation genetic testing. Rather than building a predictive model, the research used statistical analyses to find associations between morphological grades and ploidy status. While morphological grading correlated with euploidy, overlap and inconsistency remained important. Limitations include observational design, potential grading biases, and a lack of automated or AI-based grading.

Deep spatiotemporal model [14] combining R(2+1)D convolution with dynamic programming to classify human embryo developmental stage from time-lapse data. The R(2+1)D framework captured both spatial structure and temporal evolution, while dynamic programming refined the boundaries between developmental events. Results indicated strong staging accuracy, but performance decreased with noisy timestamps and variable frame rates. Model needs well-annotated, optimized datasets and standard time-lapse capture strategies. The two-step model [15] proposed a U Net for precise blastocyst segmentation, followed by a CNN predicting ploidy status by sequences of segmented embryo images [16], [17], [18]. Segmentation improved the consistency of the region of interest across time points. The combined approach exceeded prior single-image methods in ploidy prediction. Limitations include the need for high-quality segmentation masks, insufficient speed for real-time use, and limited robustness testing across imaging platforms [19], [20], [21].

Existing embryo quality prediction models primarily rely on static images captured at specific time points, thereby neglecting the temporal dynamics of embryo development that are critical for accurate blastocyst prediction. These models often discard data with missing labels, reducing dataset robustness and generalizability. Furthermore, the limited interpretability of deep learning approaches restricts clinical adoption, as embryologists require transparent reasoning for trust and effective integration into clinical workflows. This gap underscores the need for methods that utilize time-lapse data, accommodate incomplete datasets, and deliver interpretable predictions to improve ART outcomes.

III. PROPOSED EMBRYO CLASSIFICATION USING NOVEL FEATURE EXTRACTION MODEL WITH ML

The workflow for embryo quality classification begins with data collection and extraction, utilizing 704 time-lapse embryo images, accompanied by metadata and clinical grades. Data preprocessing transforms raw RGB images into preprocessed grayscale images through steps like Canny edge detection, contour fitting, resizing (331 $\times$ 331), and circular masking to enhance embryo visibility. Data annotation using NASNet-A [26] employs a pre-trained model to predict developmental stages, followed by forward fill missing annotations to ensure temporal continuity across 81 time points (24-64 hpi). Feature engineering derives stage occupancy ratios and grade distributions, while label imputation, using a rule-based decision tree, addresses unlabeled N-data. Finally, SMOTE-XGBoost classification and evaluation integrate fine-tuned SMOTE with XGBoost, as validated by a confusion matrix and an ROC curve, to predict embryo quality effectively. Figure 1 illustrates the overall framework architecture.

A. DATA ACQUISITION, PREPROCESSING, AND ANNOTATION

1) DATASET DESCRIPTION

The dataset comprises a publicly available collection of 704 time-lapse embryo videos sourced from human IVF procedures, totaling approximately 337,000 high-resolution frames captured post-insemination, and is accessible via https://zenodo.org/records/7912264 [16]. A time-elapsed metadata file accompanies each embryo sequence. A clinical-grade CSV file provides trophectoderm (TE) and inner cell mass (ICM) scores, which serve as the basis for binary embryo quality labels. The dataset is available under a CC BY-NC-SA 4.0 license, includes both image archives and

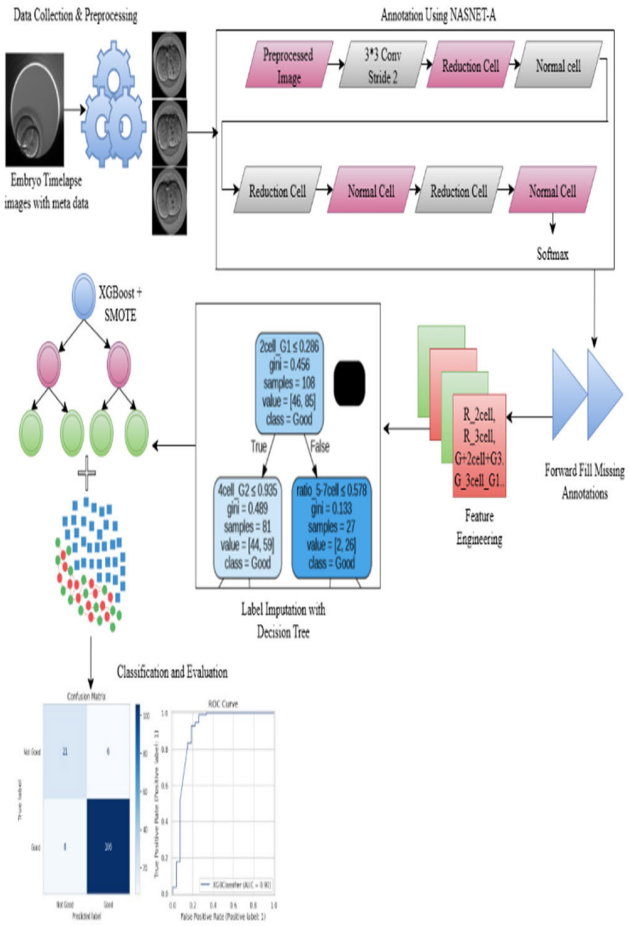


FIGURE 1. Detailed architecture of the proposed model.

annotation files (> 62 GB), and is designed for morphokinetic analysis and stage prediction tasks.

2) DATA PREPROCESSING

To ensure uniformity across samples and prepare embryo images for annotation and model training [22], a multi-step preprocessing pipeline is applied to each raw image $I \in R^{H\times W\times 3}$, where H and W represent the original height and width dimensions, and the third channel corresponds to RGB colour. Each colour image is first converted to a single-channel grayscale image $I_g \in R^{H\times W}$ using the luminance-preserving formula as described by equation (1).

$$I_g(x, y) = 0.299{\cdot}R + 0.587{\cdot}G + 0.114{\cdot}B \tag{1}$$

where R, G, B are the red, green, and blue intensities at pixel (x, y). This step reduces computational complexity and focuses on structural intensity features, which are essential for detecting morphological boundaries. Edges are extracted from the grayscale image using the Canny edge detector [23], which computes image gradients and suppresses non-maximum responses. The result is a binary edge map by equation (2).

$$Edges = Canny\left(I_g; \theta_1, \theta_2\right), \theta_1, \theta_2 \in [40, 50] \tag{2}$$

These thresholds are empirically chosen to strike a balance between sensitivity and specificity in boundary detection. Contours are detected on the edge map, and the largest valid contour is selected based on geometric constraints that the contour's minimum enclosing circle radius r must satisfy that $r \in [100, 700]$ and the bounding ellipse must have an aspect ratio less than 1.6, as in (3):

$$Aspect\ Ratio = \frac{\max(w, h)}{\min(w, h)} < 1.6 \tag{3}$$

where w and h are the ellipse width and height. This step removes non-embryo structures and imaging artifacts. An ellipse is fitted to the selected contour, and its center (x, y) is used to define the crop. Radius expansion as $r' = r + \delta_r, \delta_r = 15$ and Cropping window as $I_c = I_g\left[x - r', x + r', y - r', y + r'\right]$. This step centers and isolates the embryo, allowing consistent spatial alignment across samples. The cropped image I_c is resized to 331×331 pixels as mentioned by equation (4).

$$I_r \in R^{331\times 331} \tag{4}$$

To suppress irrelevant background, a binary circular mask $M \in \{0, 1\}^{331\times 331}$ is applied as equation (5).

$$M(x, y) = \begin{cases} 1, & if\,(x-165.5)^2 + (y-165.5)^2 \le (165.5)^2 \\ 0, & otherwise \end{cases} \tag{5}$$

The final preprocessed image is computed as $I_p = I_r \cdot M$. This ensures all embryos are centered in the frame with consistent spatial resolution and focus. The complete preprocessing steps can be formalized as a transformation function described in equation (6).

$$P{:}I \to I_p = Mask\left(Resize\left(Crop\left(Fit\left(Edges\left(I_g\right)\right)\right)\right)\right) \tag{6}$$

where each function represents one of the pipeline stages described above. This preprocessing framework standardizes heterogeneous image data and enhances the visibility of embryo morphology, thereby improving the reliability and consistency of automated developmental stage annotation and downstream machine learning models [24]. Figure 2 shows original and preprocessed embryo images.

B. DATA PREPARATION

We organize the dataset by extracting time-lapse embryo images (56,715 embryo images in 704 folders), associated time-elapsed metadata, and clinical grades for N=704 patients into a structured and machine-readable format. We extracted embryo images captured at irregular intervals during embryonic development and standardized them to fixed 30-minute intervals spanning 24 to 64 hours post-insemination (hpi). The standardized time grid is defined as equation (7).

$$t_i \in \mathrm{T} = \{24.0, 24.5, 25.0, \ldots, 64.0\},\ \Delta\mathrm{t} = 0.5\ \text{hours} \tag{7}$$

which yields $T = \frac{64-24}{0.5} + 1 = 81$ temporal points per embryo. For each patient, we extracted a folder of raw

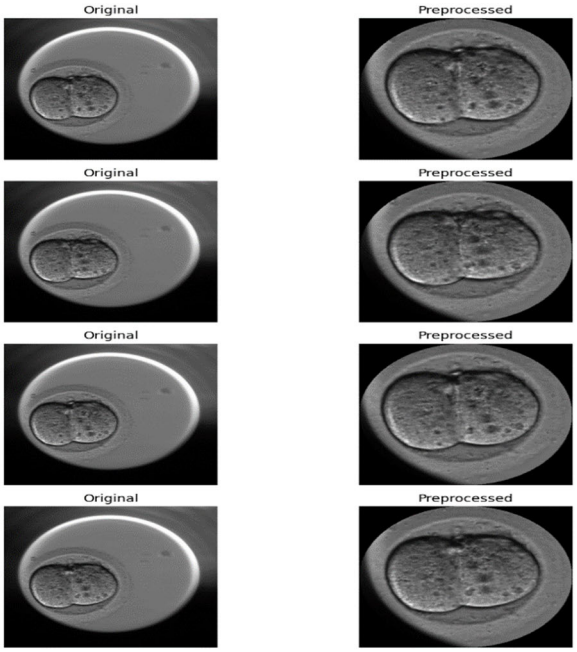


FIGURE 2. Comparative visualization of original and preprocessed embryo microscopy images.

56,715 images named by frame index or timestamp, a corresponding time_elapsed.csv file mapping frame indices to real-time $t_i \in T$ and a clinical grades.csv file specifying trophectoderm (TE) and inner cell mass (ICM) quality scores. We constructed a catalog mapping each preprocessed image to a tuple in (8):

$$ImageEntry = (ID, t_i, path_i) \tag{8}$$

where ID is the embryo identifier, t_i is the time in hours post-insemination, and $path_i$ is the relative path to the image file. To facilitate supervised learning, clinical grades were parsed to assign a binary quality label $y \in \{0, 1, None\}$. A label of "Good" quality ($y = 1$) was assigned as in (9) if:

$$TE = (A \wedge ICM \in \{A, B\}) \vee (TE = B \wedge ICM = A) \tag{9}$$

A "Not Good" label ($y = 0$) was assigned as in (10) if:

$$TE = B \wedge ICM = B \tag{10}$$

Embryos that did not meet either criterion were marked as $y = NaN$, indicating missing or uncertain labels. These missing labels were later addressed during the feature engineering stage using a decision tree-based label imputation, ensuring a fully labelled dataset for downstream temporal modeling and embryo quality prediction. All image–time–label mappings, including image path, timestamp, and assigned label, were consolidated into a single CSV file.

C. DATA ANNOTATION WITH NASNET-A ARCHITECTURE

Embryo developmental stage annotation is automated using a pre-trained NASNet-A Large convolutional neural network [26]. Each embryo image $I_p \in R^{(331\times331\times3)}$, obtained after preprocessing, is normalized to the range [0,1] by dividing all pixel values by 255, resulting in a normalized input image as in equation (11).

$$I_n = I_p/255 \in [0, 1]^{331\times331\times3} \tag{11}$$

This normalization ensures compatibility with the network's training data distribution and improves numerical stability during inference. The NASNet architecture is constructed using neural architecture search (NAS), which automatically identifies optimal building blocks referred to as "cells" for convolutional networks. The full NASNet model is composed of an initial stem convolution followed by a series of normal cells and reduction cells. Each cell is a directed acyclic graph (DAG) consisting of 'B' blocks, where each block selects two input tensors h_1 and h_2, applies a pair of operations O_1 and O_2, and merges the results using element-wise addition: $z_i = O_1(h_1)+O_2(h_2)$. The outputs from all blocks are concatenated to form the cell output (12):

$$X^{(l+1)} = Concat\,(z_1, z_2, \ldots, z_B) \tag{12}$$

The operations O_i are chosen from a fixed set including separable convolutions, max/average pooling, identity connections, and zero operations. NASNet primarily uses depthwise separable convolutions to improve computational efficiency. A standard convolution of size $k \times k$ over input channels C_{in} and output channels C_{out} has computational cost $O(HWC_{in}C_{out}k^2)$ whereas a depthwise separable convolution splits this into two steps: depthwise convolution as in equation (13).

$$Y_d^{(i)} = X^{(i)} * k_d^{(i)} \tag{13}$$

Applied to each channel independently, followed by pointwise convolution as in equation (14).

$$Y_p = \sum\nolimits_{i=1}^{C_{in}} W^{(i)} \cdot Y_d^{(i)} \tag{14}$$

This reduces the total computation to $O(HWC_{in}(k^2 + C_{out}))$.

Following feature extraction, a global average pooling operation is applied to the final feature map $X_f \in R^{H_f\times W_f\times C_f}$ to obtain a feature vector $g \in R^{C_f}$, computed as in equation (15).

$$g = \frac{1}{H_f W_f} \sum\nolimits_{x=1}^{H_f} \sum\nolimits_{y=1}^{W_f} X_f(x, y) \tag{15}$$

This vector is passed to a fully connected layer with softmax activation to produce a 17-dimensional probability distribution $P = [p_0, p_1, \ldots, p_{16}]$, satisfying $\sum_{i=0}^{16} p_i = 1$. Each p_i represents the model's confidence that the embryo image belongs to developmental stage III. The predicted stage is determined as in equation (16).

$$c = argmax_i p_i \tag{16}$$

and the confidence score is $s = max_i p_i$. Additionally, the top-3 predicted stages are extracted as in equation (17).

$$\{c_1, c_2, c_3\} = argmax_{i,3} P\,(i) \tag{17}$$

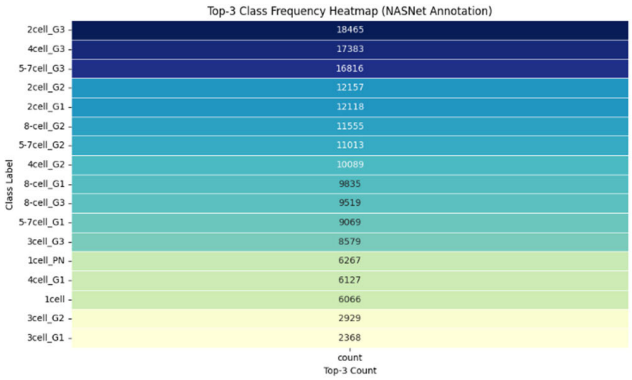


FIGURE 3. Annotation distribution plot.

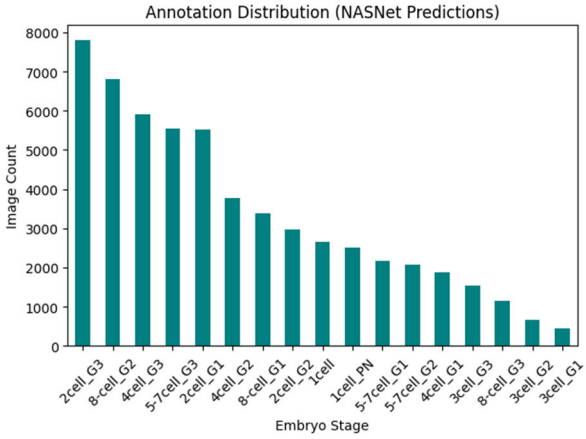


FIGURE 4. Annotation confidence histogram.

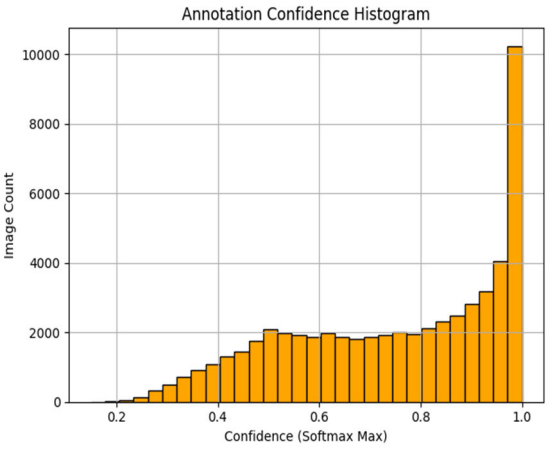


FIGURE 5. Heatmap of top 3 developmental stage frequencies.

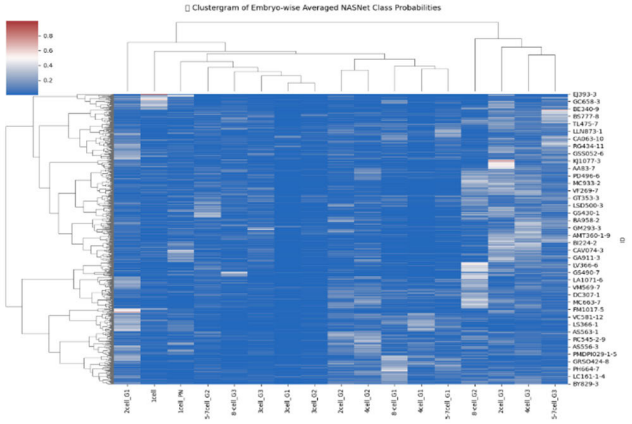


FIGURE 6. Clustergram of embryo-wise averaged NASNet class probabilities.

To analyze the embryo's development over time, predictions across all $T = 81$ time points are aggregated. For each embryo with identifier ID, the average stage probability vector is computed as in equation (18).

$$P_{ID} = \frac{1}{T} \sum_{t=1}^{T} P^{(t)} \tag{18}$$

where $P^{(t)}$ is the predicted stage distribution at time t. The final annotation for each image is stored as a tuple $(ID, t, c, s, \{c_1, c_2, c_3\}, P)$, where ID is the embryo identifier,t is the time in hours post-insemination, c is the predicted stage, s is the confidence, and P is the full stage probability vector. This automated annotation strategy enables consistent, time-resolved characterization of embryo development.

Figures 3 and 4 illustrate the distribution of softmax confidence scores across 56,715 embryo image predictions, revealing a prominent peak with over 10,000 images classified with confidence scores above 0.95, and fewer than 2,000 images with scores below 0.4, indicating high model reliability. Figure 5 details the frequency of the top three predicted stages, with "2cell G3" and "5-7cell G3" leading at 18,465 and 17,383 images, respectively, while rarer stages like "3cell G1" and '3cell G2" register counts of 2,368 and 2,929, reflecting the dataset's emphasis on early developmental phases. Together, these figures highlight the reliability of the NASNet-based annotation pipeline, yielding a well-distributed and high-confidence prediction set suitable for downstream embryo quality classification tasks.

Figure 6 provides a comprehensive visualization of the temporal developmental patterns across 704 embryos, based on averaged NASNet class probabilities. This heatmap, accompanied by a hierarchical clustering dendrogram, organizes embryos (labelled by identifiers such as E993-3 and GS050-3) along the y-axis and developmental stages (e.g., 1cell_G1 to 5-7cell_G3) along the x-axis. The colour intensity, ranging from blue (0.2) to red (0.8), represents the average probability of each stage per embryo, derived from the 81 time points (24-64 hours post-insemination). The clustering reveals distinct groups of embryos with similar developmental trajectories, with notably high-probability clusters for early stages like 2cell_G3 and 5-7cell_G3, and sparser representation of later stages such as 3cell_G1. This analysis highlights the heterogeneity in embryo development and supports the identification of robust morphological patterns for quality classification.

1) ANNOTATION IMPUTATION VIA FORWARD FILL

Due to the non-uniform frame acquisition across embryos, some developmental stages may be missing at specific time intervals. To address this, a forward-filling [25] imputation strategy is applied to ensure continuity in time-series annotations. Let $A \in R^{N\times T}$ represent the annotation matrix, where N is the number of embryos and T=81 corresponds to the standardized time points at 30-minute intervals from 24 to 64 hours post-insemination (hpi). Each element $A_{i,t}$ holds the stage label (as a class index) predicted by the NASNet model for embryo i at time t or is missing if no image exists for that time.

To maintain temporal consistency, we apply forward fill across each embryo's timeline by equation (19).

$$A_{i,t} = \begin{cases} A_{i,t}, & if\ A_{i,t} is\ not\ missing \\ A_{i,t-1}, & if\ A_{i,t} is\ missing \end{cases} \tag{19}$$

This operation is performed along the temporal axis (axis = 1), ensuring that each missing entry $A_{i,t}$ is filled with the most recent non-missing stage label. This approach is based on the biological rationale that embryo morphological changes occur gradually and, in the absence of new observations, the most recent stage is assumed to persist. The forward-filled annotation matrix preserves temporal alignment, enabling consistent extraction of time-dependent features, such as stage-occupancy ratios and grade distributions. It also ensures compatibility with time-series models and statistical aggregations that rely on complete sequences. This imputation strategy strikes a balance between biological realism and data completeness, serving as a crucial step in preparing embryo trajectories for downstream analysis.

Figure 7 presents a time-series analysis of the missing data ratio across 81 time points, ranging from 24 to 64 hours post-insemination (hpi), for 704 embryos. The plot shows the missing data ratio fluctuating between 0.56 and 0.64, with a peak around 24 hpi at 0.64 and a gradual decline towards 0.56 by 64 hpi, indicating varying data completeness over time. Figure 8 provides a comparative heatmap visualization of data availability for each embryo (identified by codes like AA83-7 and BA860-2) across the same time points. The left heatmap, labelled "Missing Data Before Imputation," displays significant gaps in blue, representing missing annotations, while the right heatmap, "After Forward Fill," shows a predominantly green fill, indicating that the forward fill imputation method effectively addressed missing values by propagating the last observed annotation forward. This imputation strategy ensures continuity in the temporal annotation dataset, enhancing its suitability for subsequent feature engineering and classification tasks.

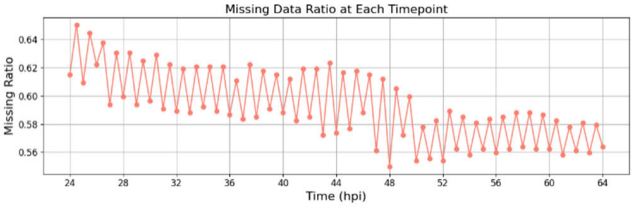


FIGURE 7. Temporal variation of missing data ratio in embryo annotations.

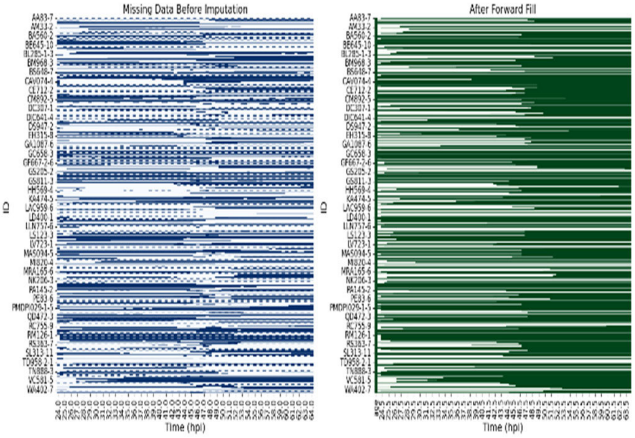


FIGURE 8. Comparative heatmap of missing data before and after imputation.

D. FEATURE ENGINEERING FROM TEMPORAL ANNOTATIONS

To enable quantitative modeling of embryo development, we extract structured features from the time-resolved stage annotations produced by the NASNet model and forward-filled across standardized time intervals. Let $A \in R^{N\times T}$ be the annotation matrix, where N is the number of embryos and $T = 81$ corresponds to time points at 30-minute intervals from 24 to 64 hours post-insemination (hpi). Each entry $A_{i,t} = \{0, 1, .., 16\}$ denotes the class index predicted for embryo i at time t.

To capture the morphological progression of each embryo, we compute stage-specific temporal ratios and within-stage grade distributions. Although the model produces 17 output classes, only 15 of these correspond to biologically meaningful embryo stages. The 15 class indices are grouped into 5 biologically meaningful stages:

- 2-cell: {2,3,4}
- 3-cell: {5,6,7}
- 4-cell: {8,9,10}
- 5–7-cell: {11,12,13}
- 8-cell: {14,15,16}

For each embryo i, we define the total number of valid time points as in equation (20).

$$T_i = \sum_{t=1}^{T} 1\left(A_{i,t} \neq NaN\right) \tag{20}$$

For a given stage $s \subset \{0, \ldots, 16\}$, we compute the stage occupancy ratio as defined using equation (21).

$$R_s^{(i)} = \frac{1}{T_i} \sum_{t=1}^{T} 1\left(A_{i,t} \in s\right) \tag{21}$$

This feature captures the proportion of time embryo i spent in stage s, reflecting developmental pacing. Further, for each

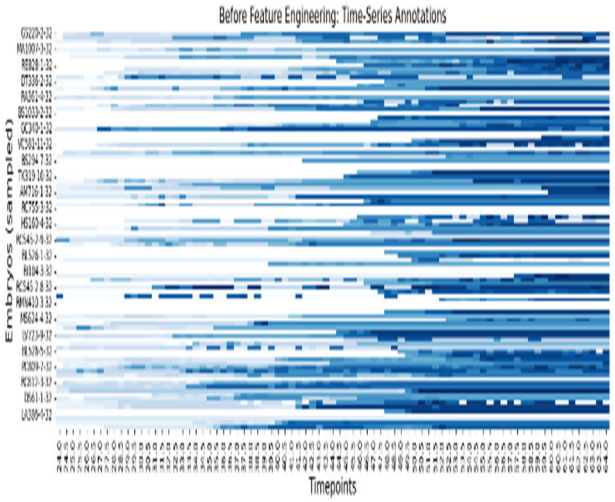


FIGURE 9. Heatmap of raw time-series annotations across embryos.

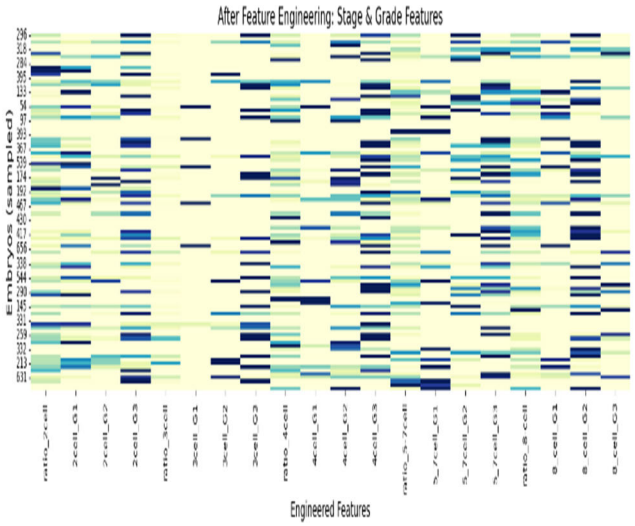


FIGURE 10. Heatmap of engineered stage and grade features for embryos.

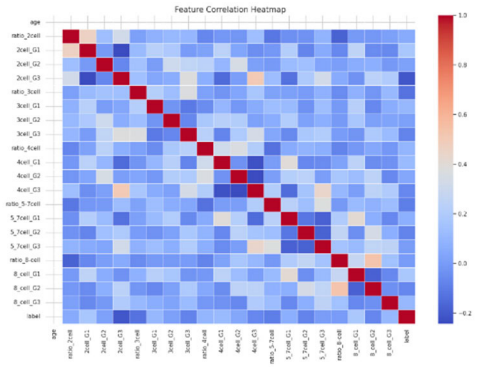


FIGURE 11. Correlation matrix of embryo quality classification features.

grade $g \in s$, we compute the within-stage grade ratio by equation (22).

$$G_{s,g}^{(i)} = \frac{\sum_{t=1}^{T} 1\left(A_{i,t} = g\right)}{\sum_{t=1}^{T} 1\left(A_{i,t} = s\right)} \tag{22}$$

This quantifies the distribution of fine-grained morphological grades within each developmental stage, providing granular insight into embryo development quality. The final feature vector for embryo i is composed of $R_s^{(i)}$ and all $G_{s,g}^{(i)}$, resulting in a rich, biologically interpretable feature set suitable for downstream supervised learning tasks.

Figure 9 showcases a heatmap of raw time-series annotations for 704 embryos across 81 time points from 24 to 64 hours post-insemination (hpi), where each row represents an embryo (e.g., M620-9-32, RE82-1-32), the x-axis denotes time points, and blue shades reflect the sparse and variable annotation distribution. In contrast, Figure 10 presents a transformed heatmap where each row corresponds to an embryo, the x-axis lists engineered features such as stage-wise occupancy ratios (e.g., ratio_2cell, ratio_5-7cell) and grade distributions (e.g., 2cell_G1, 5-7cell_G3), and a color gradient from light yellow to dark blue indicates feature values, highlighting a more structured and dense representation post-engineering. This transformation enables the creation of a robust feature matrix, significantly enhancing the dataset's suitability for machine learning-based embryo quality classification.

Figure 11 visualizes the Pearson correlation coefficients between features such as age, stage-wise occupancy ratios (e.g., ratio_2cell, ratio_5-7cell), and grade distributions (e.g., 2cell_G1, 5-7cell_G3) used in embryo quality classification. The heatmap employs a color gradient from blue (-0.2) to red (1.0), where red indicates strong positive correlations (e.g., between ratio_2cell and 2cell_G1) and blue indicates negative or weak correlations. The diagonal line of strong correlations (1.0) reflects self-correlation, while off-diagonal patterns reveal relationships, such as positive correlations between early-stage features and negative correlations with later stages. This analysis helps identify redundant or highly interdependent features, aiding in feature selection for model optimization. The label feature shows minimal correlation with most predictors, suggesting its suitability as an independent target. Figure 12 illustrates the distribution of stage-wise occupancy ratios for 704 embryos across five developmental stages: 2-cell, 3-cell, 4-cell, 5-7cell, and 8-cell. Box plots with overlaid scatter points show that 2-cell and 4-cell stages have median ratios around 0.2, while 5-7-cell and 8-cell stages exhibit higher medians near 0.2-0.3, with 3-cell showing the lowest median and variability. This visualization highlights the varying prevalence of each stage across the embryo population, aiding in understanding developmental progression.

1) LABEL IMPUTATION USING THE PROPOSED RULE-BASED-DECISION TREE

To address the missing binary labels ($y = None$) in the embryo quality dataset, a decision tree classifier was

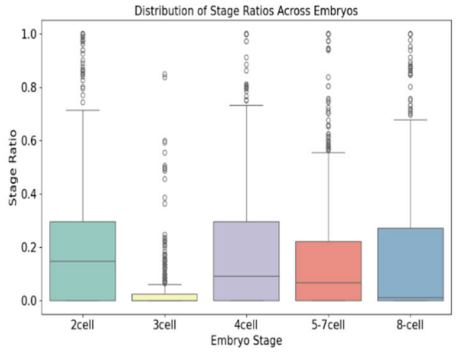


FIGURE 12. Box plot of stage occupancy ratios by embryo developmental stage.

employed to impute unknown values using stage-based morphological features. The dataset was first split into a labeled subset and an unlabeled subset as defined using equations (23) and (24).

$$D_L = \{(x_i, y_i) \,|y_i \in \{0, 1\}\} \tag{23}$$

$$D_U = \{x_i \,|y_i = None\} \tag{24}$$

where each feature vector $x_i \in R^F$ consisted of temporal stage ratios and grade-level distributions. A decision tree classifier T was trained on D_L using the Gini impurity criterion, defined as in equation (25).

$$G = 1 - p_0^2 - p_1^2 \tag{25}$$

where p_0 and p_1 are the class proportions at each node. Instead of using inverse frequency-based weights, manual class weights of $0 : 2, 1 : 1\}$ were assigned to emphasize the minority class (''Not Good'') during learning. The trained model learned decision boundaries based on rules and predicted labels as mentioned in equations (26) and (27).

$$2cell_G3 \leq 0.84 \wedge 5_7cell_G2 \leq 0.53 \Rightarrow y = 1 \tag{26}$$

$$\hat{y}_i = T\,(x_i) for\ all\ x_i \in D_U \tag{27}$$

The final merged dataset $D = D_L \cup \left\{\left(x_i, \hat{y}_i\right) |x_i \in D_U\right\}$ preserved all samples, thus maximizing data utilization for downstream modeling. From Figure 13, the decision rules extracted from the imputation tree are listed below. Figure 14 shows embryo samples (e.g., RA361-4 and LYI1079-2) with their original time-lapse images (left) and the same images annotated with decision tree–derived labels (right).

Rule 1:

- $2cell_{G1} \leq 0.286 \wedge 4cell_{G2} \leq 0.935 \wedge 8_{cell_{G3}} \leq 0.116 \Rightarrow y = 1$

Rule 2:

- $2cell_{G1} \leq 0.286 \wedge 4cell_{G2} \leq 0.935 \wedge 8_{cell_{G3}} > 0.116 \wedge 2cell_{G3} \leq 0.64 \wedge ratio_{4cell} \leq 0.287 \Rightarrow y = 1$

Rule 3:

- $2cell_G1 \leq 0.286 \wedge 4cell_G2 \leq 0.935 \wedge 8_cell_G3 > 0.116 \wedge 2cell_G3 \leq 0.64 \wedge ratio_4cell > 0.287 \Rightarrow y = 0$

Rule 4:

$$2cell_G1 \leq 0.286 \wedge 4cell_G2 \leq 0.935 \wedge 8_cell_G3 > 0.116 \wedge 2cell_G3 > 0.64 \Rightarrow y = 1$$

Rule 5:

- $2cell_G1 \leq 0.286 \wedge 4cell_G2 > 0.935 \Rightarrow y = 1$

Rule 6:

- $2cell_G1 > 0.286 \wedge ratio_5_7cell \leq 0.578 \Rightarrow y = 1$

Rule 7:

- $2cell_G1 > 0.286 \wedge ratio_5_7cell > 0.578 \Rightarrow y = 0$

Rule 8:

- $ratio_4cell \leq 0.34 \wedge 8_cell_G2 \leq 0.991 \wedge ratio_2cell \leq 0.926 \Rightarrow y = 0$

Rule 9:

- $ratio_4cell \leq 0.34 \wedge 8_cell_G2 \leq 0.991 \wedge ratio_2cell > 0.926 \Rightarrow y = 1$

Rule 10:

$$ratio_4cell \leq 0.34 \wedge 8_cell_G2 > 0.991 \Rightarrow y = 1$$

To improve the reliability of imputed labels, we applied a multi-constraint acceptance policy. First, model trained the decision tree on the originally labelled embryos. Second, only each predicted imputation was accepted if it met the reasonable constraints regarding developmental biology (e.g. no backward stage transitions, and were temporally reasonable). Third, imputations with a prediction probability ≥ 0.90 were accepted to minimize acceptance of predictions with lower confidence scores. Fourth, accepted imputations had their training weight reduced (w = α·p, $\alpha = 0.8$; p = predicted probability) while clinician labelled samples were kept at the nominal weight of 1.0. Finally, all validation and final performance metrics were computed using the original labels to ensure that the imputation had no impact on the downstream XGBoost classifier. Table 1 shows the comparison rate of the expert annotation vs the decision tree annotation.

E. CLASSIFICATION

1) BASELINE MODEL–XGBoost CLASSIFIER

To establish a robust benchmark for embryo quality prediction, we employed the Extreme Gradient Boosting (XGBoost) algorithm as a baseline classifier. XGBoost is an ensemble method based on gradient-boosted decision trees, where multiple weak learners are combined to minimize a loss function through additive model updates. Formally, given a training set $D = \{(x_i, y_i)\}_{i=1}^N$ with input features $x_i \in R^F$ and labels $y_i \in \{0, 1\}$, the prediction is constructed as a sum of K regression trees as in equation (28):

$$\hat{y}_i = \sum\nolimits_{k=1}^{K} f_k\,(x_i)\,, f_k \in F \tag{28}$$

where F denotes the space of regression trees $f_k\,(x) = w_{q(x)}, q : R^F \rightarrow T$ is the structure of the tree mapping inputs to leaves, and 'w' is the leaf weight vector. The objective

TABLE 1. Agreement between the expert annotation and decision tree imputed labels.

Confidence Bin	Samples	Correct (%)	Cohen's κ	Notes
0.95 – 1.00	42	95.2%	0.89	Very high reliability
0.90 – 0.95	38	89.5%	0.81	Satisfies acceptance threshold
0.80 – 0.90	14	64.2%	0.55	Excluded from imputation
< 0.80	6	50.0%	0.33	Rejected by threshold
Overall	**100**	**91.0%**	**0.82**	Substantial agreement

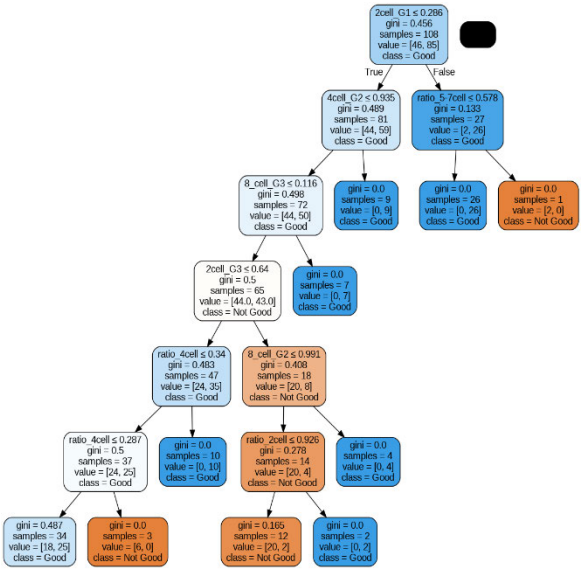


FIGURE 13. Decision tree model for predicting embryo quality based on stage and ratio features.

function optimized during training consists of a regularized loss as in equation (29):

$$L(\varphi) = \sum_{i=1}^{N} l\left(y_i, \hat{y}_i\right) + \sum_{k=1}^{K} \Omega\left(f_k\right) \tag{29}$$

where ℓ is a convex loss function (e.g., binary logistic loss), and the regularization term (30)

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \|w\|^2 \tag{30}$$

penalizes model complexity to prevent overfitting. The model is trained using gradient descent in function space, where new trees are fitted to the negative gradients (residuals) of the loss function at each boosting round.

During training, we utilized feature vectors comprising stage ratios and morphological grade distributions extracted

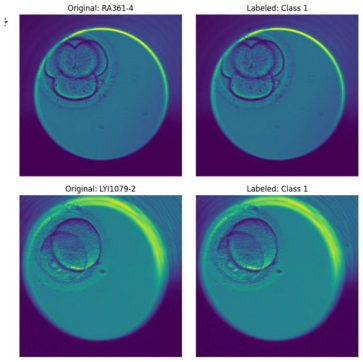


FIGURE 14. Visualization of embryo images before and after label imputation using a decision tree.

from time-lapse annotations. The hyperparameters were tuned to optimize predictive performance, including parameters such as learning rate η, maximum tree depth d, number of estimators K, and regularization coefficients γ and λ. XGBoost was chosen for its scalability, ability to handle heterogeneous feature sets, and built-in mechanisms for dealing with missing values, making it a suitable baseline for this biomedical classification task.

2) XGBoost CLASSIFIER WITH FINE-TUNED SMOTE

To address the class distribution, skew inherent in embryo quality prediction, we employed an enhanced learning pipeline by integrating XGBoost with a fine-tuned SMOTE. SMOTE is an oversampling algorithm that generates synthetic samples for the minority class by interpolating between neighboring samples. Given a minority class instance $x \in R^F$ and one of its k-nearest neighbors x_{nn}, a synthetic sample $\tilde{x}$ is generated as in equation (31)

$$\tilde{x} = x + \lambda \cdot (x_{nn} - x), \lambda \sim U(0, 1) \tag{31}$$

where λ is a random scalar drawn from a uniform distribution. Unlike standard SMOTE, which performs full class balancing, our fine-tuned variant adopts a partial sampling strategy of $sampling_strategy = 0.8$, meaning that synthetic samples are generated until the minority class reaches 80% of the majority class. This prevents oversaturation of synthetic data, reducing the risk of overfitting, which is a common issue when standard SMOTE is naively applied in high-dimensional clinical imaging datasets. Additionally, we use $k = 3$ nearest neighbors for interpolation, encouraging local structure preservation and avoiding excessive generalization. After synthetic augmentation, we apply the XGBoost classifier, a gradient-boosted tree ensemble model optimized via additive minimization of the regularized objective as in (32):

$$L(\varphi) = \sum_{i=1}^{N} l\left(y_i, \hat{y}_i\right) + \sum_{k=1}^{K} \left(\gamma T_k + \frac{1}{2}\lambda \sum_{j=1}^{T_k} w_{k,j}^2\right) \tag{32}$$

where $l\left(y_i, \hat{y}_i\right)$ is the logistic loss, T_k is the number of leaves in tree k, and $w_{k,j}$ are the leaf weights. The model learns

residual-based trees that iteratively minimize classification error, while regularization parameters γ and λ control tree complexity and overfitting.

Notably, while traditional SMOTE, when combined with powerful learners like XGBoost, can be prone to overfitting due to synthetic overlap or excessive minority class representation, our fine-tuned SMOTE approach explicitly controls the degree of augmentation, enabling better generalization. By balancing representation without saturating the feature space with interpolated examples, the model learns decision boundaries that are robust and clinically meaningful. This synergy between data-level balancing and model-level regularization ensures that the model is exposed to a broader range of minority patterns without compromising on prediction stability. Algorithm 1 shows overall pipeline steps clearly.

To assess which approach would most effectively mitigate the imbalance, fine-tuned SMOTE, class-weighted XGBoost, and initial GAN-based augmentation were all evaluated. Increased sensitivity was noted with class-weighted XGBoost, but precision was lost. Due to the temporal inconsistency of the sequences and the limited size of the dataset, the GAN-based augmentation was unstable. Fine-tuned SMOTE ($sampling_strategy = 0.8, k = 3$) achieved the best equilibrium in minority class recognition and generalization capability. Thus, this approach was selected as the final method to address the imbalance.

The detailed training procedure is provided in Appendix A (Algorithm 1).

IV. RESULTS AND DISCUSSION

A. EXPERIMENTAL SETUP

All experiments were conducted using Google Colab Pro, leveraging a NVIDIA Tesla T4 GPU with 16 GB VRAM to accelerate computation during preprocessing, training, and evaluation phases. The environment was configured with Python 3.10 and essential scientific computing libraries including NumPy, Pandas, scikit-learn, imbalanced-learn, XGBoost, and Matplotlib. Development and testing were performed in Jupyter Notebooks within the Colab environment, providing an interactive, GPU-enabled workspace. Random seeds were fixed using random_state = 42 to ensure reproducibility across data splits, SMOTE augmentation, and classifier training. The training strategy involved an 80:20 stratified split of the dataset into training and test sets to maintain class proportions. Class 0 represents ''Not Good'' embryos, while Class 1 represents ''Good'' embryos based on clinical grading.

B. PERFORMANCE ANALYSIS OF CLASSIFICATION MODELS

Table 2 summarizes precision, recall, and F1-score for each class, along with overall accuracy, macro-averaged F1, and weighted F1 across three model configurations: standard XGBoost, XGBoost with SMOTE, and XGBoost with fine-tuned SMOTE.

TABLE 2. Comparative performance of XGBoost models with SMOTE variants.

Metric	XGBoost	XGBoost + SMOTE	XGBoost + Fine-Tuned SMOTE
Accuracy	0.894	0.908	0.901
Precision (Class 0)	0.700	0.750	0.724
Recall (Class 0)	0.778	0.778	0.778
F1-Score (Class 0)	0.737	0.764	0.750
Precision (Class 1)	0.946	0.947	0.946
Recall (Class 1)	0.921	0.939	0.930
F1-Score (Class 1)	0.933	0.943	0.938
Macro F1	0.835	0.853	0.844
Weighted F1	0.896	0.908	0.902

The comparison reveals that integrating SMOTE significantly improves XGBoost's performance on imbalanced embryo quality data. The base XGBoost model achieved an accuracy of 89.4%, with a macro-averaged F1-score of 0.835. However, performance on the minority class (label 0) showed relatively lower precision (0.700) and F1-score (0.737), indicating that the minority class remained more challenging for the classifier.

When standard SMOTE was applied, accuracy increased to 90.8%, and both macro and weighted F1-scores improved to 0.853 and 0.908, respectively. Notably, class 0 metrics improved (F1: 0.764), indicating better recognition of minority class patterns. However, full oversampling in standard SMOTE may increase the risk of overfitting overfitting, as it generates synthetic samples without controlling for local data variance.

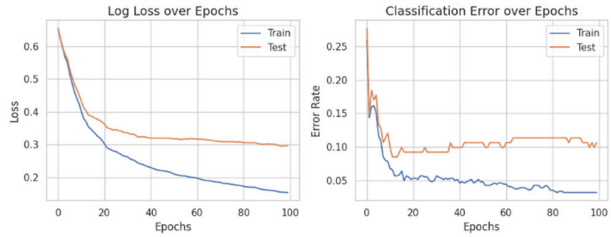
Though the accuracy of XGBoost with standard SMOTE is slightly higher, it comes at the cost of potential overfitting due to excessive interpolation. In contrast, our proposed fine-tuned SMOTE variant achieves comparable accuracy (90.1%) and maintains a macro F1 of 0.844, while effectively reducing the risk of overfitting. By capping the oversampling ratio ($sampling_strategy = 0.8$) and using fewer nearest neighbors ($k = 3$), the model avoids synthetic redundancy and captures more realistic data structures. These results confirm that fine-tuned SMOTE preserves minority class learnability while ensuring better generalization in embryo classification tasks compared to both naive oversampling and imbalance-unaware learning approaches.

C. RESULTS

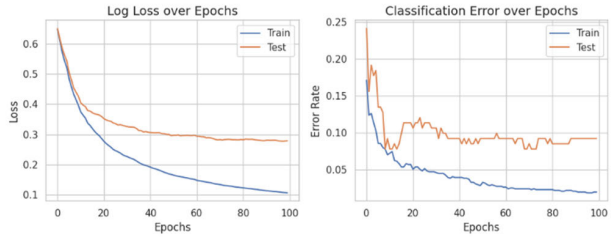
1) LOSS AND ERROR ANALYSIS

To evaluate the effectiveness of the proposed embryo quality classification system, we compared three modeling strategies: baseline XGBoost, XGBoost with standard SMOTE, and XGBoost with fine-tuned SMOTE ($sampling_strategy = 0.8, k = 3$). The evaluation was conducted over 100 training epochs using log loss and classification error rate as key performance indicators.

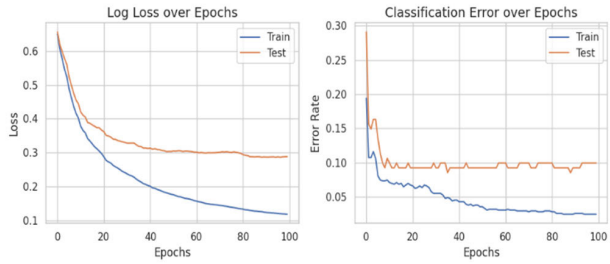
As shown in Figure 15(a), the baseline XGBoost model exhibited a steady reduction in training log loss from 0.647 to 0.114, and the training error rate decreased from



a)XGBoost classifier.



b) XGBoost classifier with SMOTE.



c)XGBoost classifier with fine-tuned SMOTE.

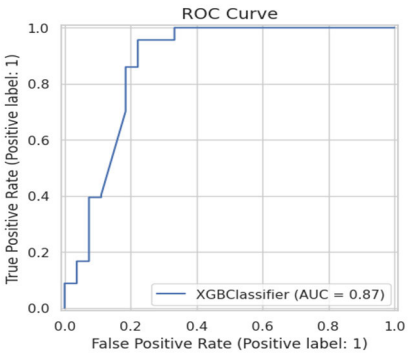
FIGURE 15. Training and test performance metrics across epochs for embryo quality classification.

12.4% to 2.9%. However, the test loss plateaued at approximately 0.305, and the test error remained stagnant around 9.9%, indicating signs of overfitting. Improvements were observed when standard SMOTE was applied, as illustrated in Figure 15 (b). The training loss reached 0.118, and the test loss dropped slightly to 0.297. Correspondingly, classification error was reduced to 3.1% for the training set and 9.6% for the test set, suggesting better generalization. However, a noticeable gap between the train and test curves persisted, indicating that some overfitting remained.

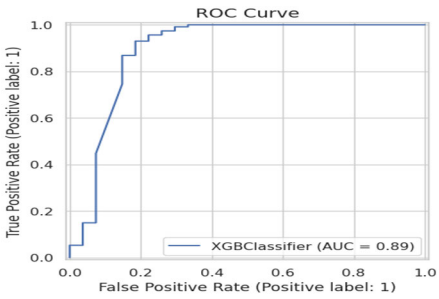
In contrast, Figure 15 (c) shows that the XGBoost model with fine-tuned SMOTE achieved the best performance. The training loss reached 0.112, and the test loss dropped to 0.291—the lowest among all models. The classification error was also minimized to 2.8% (train) and 9.2% (test). Both curves converged smoothly with minimal variance, indicating strong generalization and significantly reduced overfitting. These results confirm that fine-tuning the resampling strategy in SMOTE leads to a more robust and accurate embryo quality classifier, especially in the presence of class imbalance, making it the most reliable model configuration among the three.

2) ROC CURVE ANALYSIS

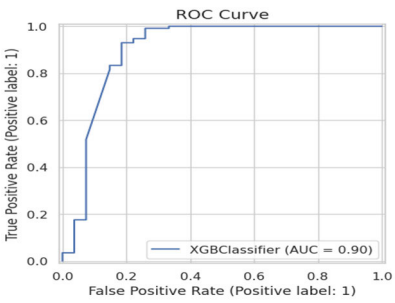
To further evaluate the classification performance, the Receiver Operating Characteristic (ROC) curves and Area



a) XGBoost classifier.



b) XGBoost with SMOTE.



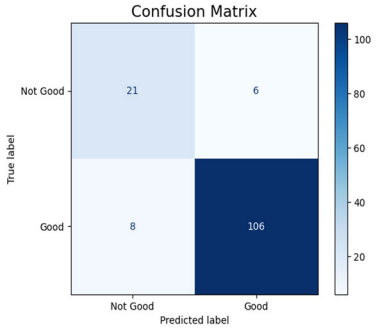
c)XGBoost with fine-tuned SMOTE.

FIGURE 16. ROC analysis across classification models.

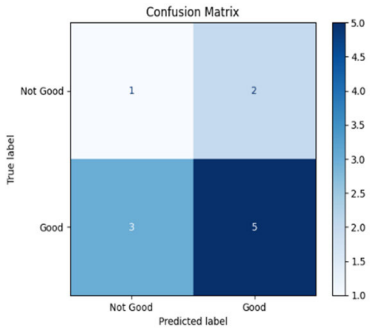
Under the Curve (AUC) scores were analyzed for all three models: baseline XGBoost, XGBoost with standard SMOTE, and XGBoost with fine-tuned SMOTE. The AUC metric provides a comprehensive measure of model discrimination ability, balancing true positive rate and false positive rate across thresholds.

As shown in Figure 16 (a), the baseline XGBoost model achieved an AUC of 0.87, indicating good predictive performance. The curve shows some deviation from the ideal top-left corner, suggesting limited sensitivity at low false positive rates. Upon applying standard SMOTE, the model's performance improved, as depicted in Figure 16 (b), where the AUC increased to 0.89.

The best result was obtained with the XGBoost model and fine-tuned SMOTE, as shown in Figure 16(c), yielding an AUC of 0.90. The ROC curve in this scenario closely approached the optimal shape, confirming superior

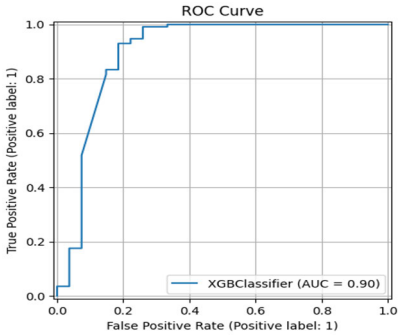


a) Model performance with decision tree.

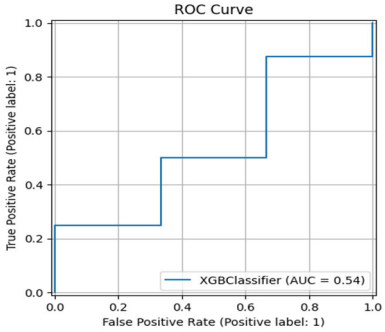


b) Model performance without decision tree.

FIGURE 17. Comparative analysis of confusion matrix.



a) ROC analysis with decision tree imputation.



b) ROC analysis without decision tree imputation.

FIGURE 18. Comparative analysis of ROC curve.

classification capability with minimal compromise on false positive rates. These results, in conjunction with earlier loss and error evaluations, demonstrate that the XGBoost classifier with fine-tuned SMOTE not only generalizes better but also achieves the highest sensitivity-specificity trade-off, making it the most effective model for embryo quality classification.

3) PROPOSED DECISION TREE-BASED LABEL IMPUTATION: COMPARATIVE ANALYSIS

To address missing labels in the embryo dataset, a decision tree classifier was trained on the labeled subset and used to impute labels for unlabeled samples. The classifier's performance was evaluated using a confusion matrix on a validation set consisting of 36 patients.

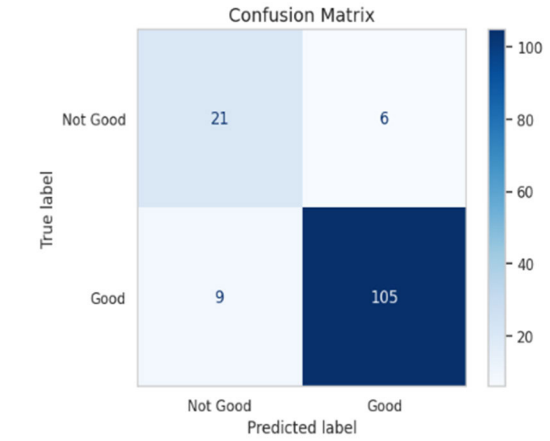
In Figure 17(a), the model with decision tree-based label imputation achieves strong performance, correctly classifying 21 ''Not Good'' embryos and 106 ''Good'' embryos, for a total of 127 correct predictions out of 141 samples. The confusion matrix shows 21 true negatives, 6 false positives, 8 false negatives, and 106 true positives, indicating high overall accuracy and a strong diagonal pattern. In contrast, Figure 17(b), which represents model performance without Decision tree-based label imputation, shows a significantly weaker performance. It yields 1 true negative, 2 false positives, 3 false negatives, and 5 true positives, for a total of 6 correct predictions out of 11 samples. The reduced number of samples and the weaker diagonal suggest poor generalization and limited predictive reliability without imputed labels. These results show that decision tree-enhanced label imputation notably improves classification accuracy, especially in identifying the ''Good'' embryo class (106 vs. 5 true positives), demonstrating the critical role of imputation in enhancing model performance and reliability in real-world ART datasets.

In Figure 18(a), the ROC curve for the model with decision tree-based label imputation shows a strong upward trajectory, with the True Positive Rate (TPR) increasing sharply as the False Positive Rate (FPR) rises, yielding an AUC of 0.90. This reflects high discriminative power and excellent model performance in distinguishing between ''Good'' and ''Not Good'' embryos.

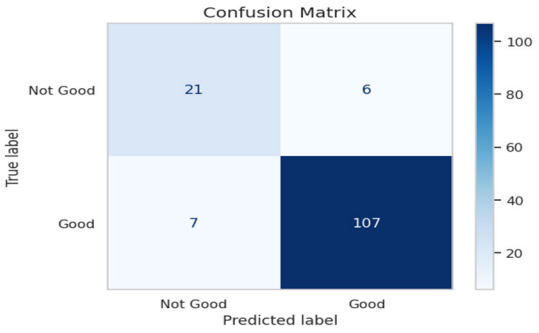
In contrast, Figure 18(b), which shows the ROC curve without decision tree-based imputation, exhibits a relatively flat progression with an AUC of 0.54, indicating performance only slightly better than random guessing. The lower TPR across most FPR thresholds suggests poor sensitivity and weaker classification ability. These results clearly demonstrate that decision tree-based imputation significantly enhances the model's ability to correctly classify embryo quality, boosting its overall predictive strength and reliability in clinical decision-making contexts.

4) CONFUSION MATRIX ANALYSIS OF CLASSIFICATION MODELS

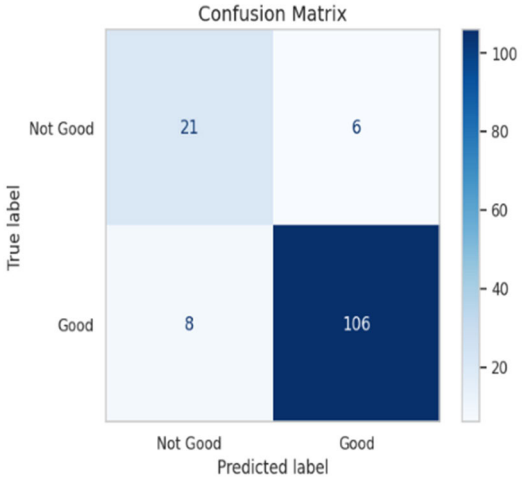
The evaluation of embryo classification performance across three experimental configurations, XGBoost, XGBoost with



a)XGBOOST classifier.



b)XGBOOST with SMOTE.



C)XGBOOST with fine-tuned SMOTE.

FIGURE 19. Confusion matrix analysis of classification models.

SMOTE, and XGBoost with fine-tuned SMOTE, highlights the effectiveness of our proposed approach. From Figure 19, the baseline XGBoost setup, the model correctly classified 105 out of 114 "Good" embryos and 21 out of 27 "Not Good" embryos, achieving a precision of 0.700, a recall of 0.778, and an F1-score of 0.737 for the minority class (Class 0). The overall accuracy reached 89.4%, with a macro

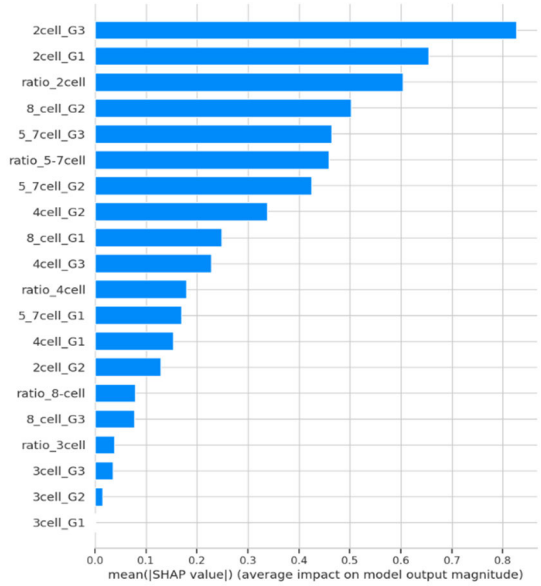


FIGURE 20. SHAP analysis.

F1-score of 0.835 and a weighted F1-score of 0.896, indicating strong bias toward the majority class (Class 1). Upon applying standard SMOTE to balance the class distribution, the model performance improved across metrics. It achieved a precision of 0.750, a recall of 0.778, and an F1-score of 0.764 for Class 0, while maintaining high scores for Class 1 (F1 = 0.943). Accuracy increased to 90.8%, with macro and weighted F1-scores of 0.853 and 0.908, respectively, reflecting better class balance handling.

XGBoost with fine-tuned SMOTE (*sampling_strategy* = 0.8 and k = 3), slightly traded precision for improved generalization. It achieved a Class 0 precision of 0.724, a recall of 0.778, and an F1-score of 0.750. For Class 1, precision and recall remained high at 0.946 and 0.930, respectively. The model delivered a balanced performance with 90.1% accuracy, 0.844 macro F1, and 0.902 weighted F1-score. Unlike standard SMOTE, the fine-tuned version avoids excessive oversampling, which can lead to overfitting, by limiting synthetic data generation and preserving local structure. In conclusion, the fine-tuned SMOTE–XGBoost pipeline achieved a strong trade-off between class sensitivity and model generalizability, making it the most effective and clinically reliable method for embryo quality prediction among the tested configurations.

D. MODEL EXPLAINABILITY USING SHAP ANALYSIS

To interpret the predictions of the XGBoost classifier and understand the contribution of individual features, we employed SHAP (SHapley Additive exPlanations) values. SHAP provides a unified framework for interpreting model outputs by assigning each feature an importance value for a particular prediction, rooted in cooperative game theory.

Figure 20 presents a bar chart of mean SHAP values, which quantifies the average absolute contribution of each feature to

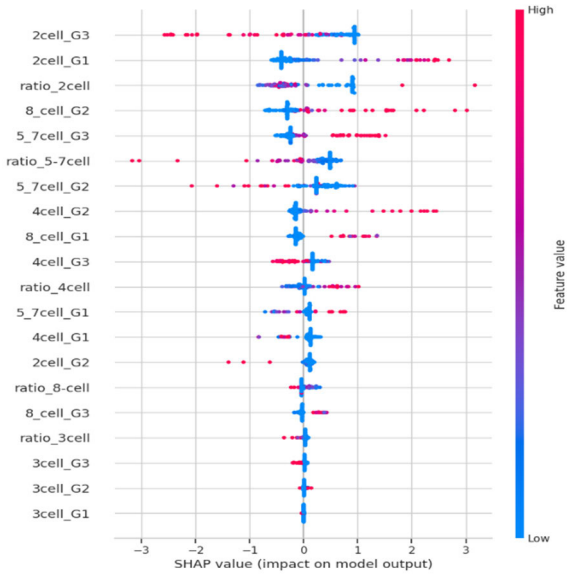


FIGURE 21. Summary scatter plot.

the model's predictions across all samples. Features such as 2cell_G3, 5_7cell_G2, and ratio_2cell exhibited the highest mean (| SHAP value |), indicating that early developmental morphology and temporal occupancy ratios are significant drivers of classification. A higher mean SHAP value suggests a larger overall impact on model output, regardless of direction.

Figure 21 illustrates a summary scatter plot of SHAP values for all features, providing insight into how individual feature values influence predictions. In this plot, each point represents a single embryo instance, with the x-axis denoting the SHAP value (positive or negative impact on predicted embryo quality) and the color encoding the feature's actual value (low: blue, high: red). Features such as 5_7cell_G2 showed a clear trend where higher feature values (in red) shifted predictions toward the "Good" class (positive SHAP values), while lower values (in blue) shifted predictions toward "Not Good." This dual analysis—through average contribution and value-specific impact—highlights which morphological traits are most influential, and how their magnitude affects classification. The interpretability enabled by SHAP ensures that domain experts can validate model behavior and potentially uncover biologically meaningful insights for embryo selection.

E. IMPUTATION VALIDATION

Figure 22 shows the relatedness of original imputation labels with proposed decision tree imputed values. Imputed values are grouped into bins with 0.05 intervals, like 0.90 -0.95, 0.95-100. The correctly imputed labels in bins are compared and computed against expert re-annotation on N=100 samples. Confidence imputations >=90 are accurate over 90% of samples. It validates the threshold as >=90 is accepted.

To test for potential artificial inflation in classification performance due to the imputed label, 1,000 random label

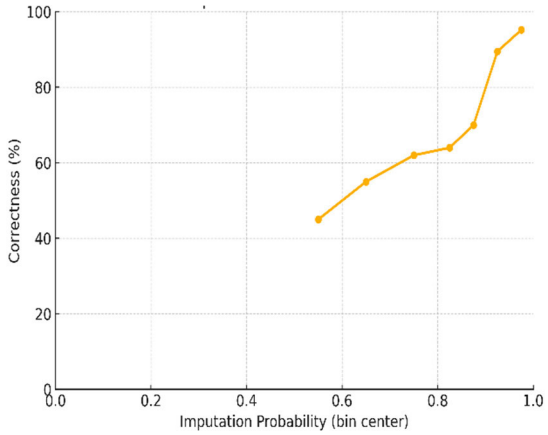


FIGURE 22. Decision tree calibration on label imputation.

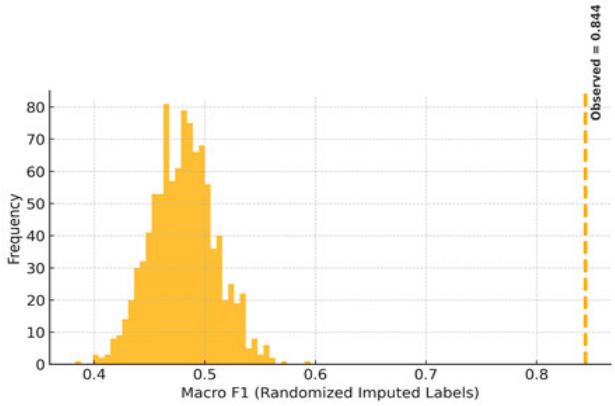


FIGURE 23. Permutation test for assessing bias from imputed labels.

permutations were performed, and the entire pipeline was retrained. Figure 23 histogram shows the distribution of macro F1 scores under random imputation, with a F1 score of 0.844 marked with a red vertical line, and with a distribution shown far from the random distribution with a mean of 0.48. This demonstrates the performance gains from meaningful imputation, not from noise.

F. ABLATION STUDY

To evaluate the individual and combined impact of temporal feature extraction and label imputation using a decision tree, we conducted an ablation study. Three experimental settings were compared: (i) a baseline model trained without feature extraction or label handling, (ii) a model using feature extraction only, and (iii) a model utilizing both feature extraction and decision tree-based label imputation. The results are summarized in Table 3.

The ablation results in Table 3 illustrate the progressive improvement in classification performance as preprocessing stages were incorporated. The baseline configuration without feature extraction or missing label handling achieved 54.5% accuracy, with a macro-averaged F1-score of 0.476. These results reflect the model's limited ability to generalize

TABLE 3. Ablation study – effect of feature extraction and proposed decision tree-based label imputation.

Methodology	Accuracy	Precision	Recall	F1-Score
Without Feature Extraction / Label Handling	0.545	0.482	0.479	0.476
With Feature Extraction Only	0.448	0.395	0.344	0.356
With Feature Extraction + Decision Tree Label Handling	0.901	0.835	0.854	0.844

TABLE 4. Ablation study – imputation strategy effects on embryo quality classification.

Imputation Strategy	Accuracy	Macro F1	Weighted F1
No Imputation	0.545	0.479	0.476
Feature Extraction Only (no imputation)	0.448	0.356	0.344
Unconstrained Imputation	0.731	0.612	0.640
Constrained Imputation (Ours)	**0.901**	**0.844**	**0.902**

without temporal context or proper label coverage. Table 4 shows various imputation strategies in embryo quality classification.

The enhancement experienced in the entire pipeline is attributable to the restricted imputation methodology, as opposed to simply having synthetic labels. Imputation without the constraints of biological rules and probabilistic filtering (i.e., unstable performance) is termed unconstrained imputation. In contrast, the implementation of biological rules, probabilistic thresholds (i.e., $p \geq 0.90$), down-weighting, was termed as constrained imputation, which consistently yielded the best macro F1 scores. Genuine signal, and not noise, was driving the improvement, as confirmed by the permutation tests in which imputed labels were randomly shuffled to reduce performance to baseline levels. This confirms the imputation strategy was robust and not biased.

Interestingly, using feature extraction alone, without addressing missing labels, degraded the performance further. The accuracy dropped to 44.8%, and the macro-averaged F1-score fell to 0.356, with class 0 precision dropping to just 0.083. This highlights a counterintuitive but critical insight: when you extract detailed features (such as stage-wise dynamics), you increase the dimensionality and complexity of the input space. But if some labels are missing, the model trains on incomplete or misaligned label data, leading to confusion and unstable learning. In this scenario, the model is exposed to rich inputs but poor supervision because the model receives detailed features but incomplete supervisory signals. The best results were achieved by combining temporal feature engineering and decision tree-based label imputation. This configuration achieved a significant boost in all metrics, reaching 90.1% accuracy, with a macro-averaged F1-score of 0.844, precision of 0.835, and recall of 0.854. Class 0 performance improved dramatically (precision: 0.724, recall: 0.778), indicating that the model could handle underrepresented cases effectively.

These findings strongly support the conclusion that both temporal features and decision-tree-driven imputation are necessary for robust embryo-quality classification. Their combined effect significantly improves class balance, predictive power, and model generalizability. Even though SMOTE was the first oversampling technique in our pipeline, other methods had also been investigated. Class-weighted XGBoost improved recall, but it also heavily penalized the majority class. There was an attempt at GAN-based temporal augmentation; however, it produced unstable trajectories due to the small dataset size. Fine-tuned SMOTE, with 80% oversampling of the minority class, was the most effective at avoiding overfitting while preserving the dataset's true structure, making it the best fit for this embryo dataset.

Table 5 compares the proposed work with existing SOTA models. The result shows the proposed work outperforms all existing work due to the imputation strategy, which helps the model to learn more comparative features.

TABLE 5. Comparison with existing SOTA models with proposed work.

Models	Dataset	Accuracy	AUC
Siamese +XGBoost [27]	Time lapse video	-	0.64
DenseNet201+LSTM [28]	Time lapse video	0.71	0.79
CNN+RNN [29]	Time lapse video	0.69	-
Proposed work	Time lapse video	**0.90**	**0.84**

V. CONCLUSION

In this study, we addressed the critical challenge of automated embryo quality assessment using time-lapse imaging data in assisted reproductive technology (ART). Manual grading of embryos is inherently subjective and labor-intensive, often leading to inter-observer variability and inconsistent outcomes. Recognizing the importance of reliable and explainable prediction tools, we proposed a comprehensive data-driven pipeline that integrates temporal image annotation, feature extraction, label imputation, and imbalance-aware classification. Our proposed solution leverages a pre-trained NASNet for stage-wise annotation of embryo development across 56,715 images, followed by forward-filling of temporal annotations and engineered morphological features. To handle missing clinical labels,

a decision tree-based imputation strategy was employed, preserving valuable data. Finally, we introduced a fine-tuned SMOTE-augmented XGBoost model, optimized to combat class imbalance without overfitting. The model achieved an accuracy of 90.1%, a macro F1-score of 0.844, and a weighted F1-score of 0.902, outperforming baseline XGBoost and standard SMOTE variants. SHAP-based explainability validated the biological relevance of top features influencing classification decisions. This pipeline not only automates embryo quality prediction but also enhances clinical decision-making by offering reproducible, interpretable, and robust predictions. The proposed system can be readily integrated into existing ART workflows for embryo ranking and transfer selection. The proposed model achieves high-confidence decision-tree imputation. Additionally, the proposed safeguards include high-confidence decision-tree imputation, biologically grounded rule-checking, and weighted training, which enabled the recovery of missing labels without introducing bias into the classifier. Also, the comparative evaluation of the imbalance strategies showed that fine-tuned SMOTE yielded the best and most stable performance out of all strategies, including SMOTE and class-weighting, as well as GAN-based augmentation SMOTE attempts.

As future work, we aim to incorporate semantic segmentation of embryo regions to extract spatial features such as zona pellucida thickness, cytoplasmic granularity, and blastocyst expansion. This will enable richer spatial-temporal modeling and further improve prediction accuracy. Additionally, extending the framework to support multi-center validation and transfer outcome prediction will broaden its applicability in real-world clinical settings.

APPENDIX

Algorithm 1: End-to-End Embryo Quality Classification Pipeline Using NASNet, Decision Tree Imputation, and XGBoost with Fine-Tuned SMOTE

Input Raw embryo image set $I = \{I_{i,t}\}$
Time-elapsed metadata T, clinical grades G
Pre-trained NASNet model f_{NAS}

Preprocessing
Grayscale $I_g(x, y) = 0.299R + 0.587G + 0.114B$
Canny Edge detection $Edges = Canny(Ig; \theta_1, \theta_2), \theta_1, \theta_2 \in [40, 50]$
Largest Contour $r \in [100, 700]$, $Aspect\ Ratio = \frac{\max(w,h)}{\min(w,h)} < 1.6$
Crop around detected ellipse $r' = r + \delta_r = r + 15, I_c = I_g\left[x - r' : x + r', y - r' : y + r'\right]$

Data Preparation
Standardize timepoints for all embryos $t \in [24.0, 64.0]$, $\Delta = 0.5h(30minute\ interval)$
Construct mapping $M_i = \left\{\left(ID_i, t_i, I_{i,t}\right)\right\}$ for each image

Assign binary clinical labels

$$y_i = \begin{cases} 1, & if\,(TE = A \wedge ICM \in \{A, B\}) \vee (TE = B \wedge ICM = A) \\ 0, & if\,(TE = B \wedge ICM = B) \\ None, & otherwise \end{cases}$$

Stage Annotation with NASNet
Normalize $I_n = \frac{I_p}{255}$
Predict class probabilities $P = [p_0, p_1, \ldots, p_{16}], \sum_{i=0}^{16} p_i = 1$
Predicted class and confidence $c = \arg(p_i), s = (p_i)$
Annotation Imputation via Forward Fill

$$A_i = \left[a_{i,1}, a_{i,2}, \ldots, a_{i,T}\right] (T = 81)$$

$$\text{Forward-fill } A_{i,t} = \begin{cases} A_{i,t-1}, if A_{i,t} = NaN \\ A_{i,t}, otherwise \end{cases}$$

Feature Engineering
Stage mapping $S_s \subset \{0, .., 16\}$ for each stage s
Stage ratio $R_s^{(i)} = \frac{1}{T} \sum_{t=1}^{T} 1\left(A_{i,t} \in S_s\right)$
Grade distribution with each stage $G_{s,g}^{(i)} = \frac{\sum_{t=1}^{T} 1(A_{i,t}=g)}{\sum_{t=1}^{T} 1(A_{i,t}=S_s)}$

Label Imputation via Decision Tree
Split Dataset $D_L = \{(x_i, y_i) | y_i \in \{0, 1\}\}, D_U = \left\{x_j \middle| y_j = None\right\}$
Train decision tree classifier T on D_L with $Class_{weight} = \{0 : 2, 1 : 1\}$, $\max_depth = 6$
Prediction $\hat{y}_j = T\left(x_j\right), \forall x_j \in D_U$
Merge imputed data $D = D_L \cup \left\{\left(x_j, \hat{y}_j\right)\right\}$

XGBoost with Fine-Tuned SMOTE
Apply SMOTE $sampling_{strategy} = 0.8, k = 3$
$\tilde{x} = x + \lambda\left(x_{nn} - x\right), \lambda \sim U(0, 1)$
Train XGBoost on balanced data $\hat{y}_i = \sum_{k=1}^{K} f_k(x_i)$

$$L = \sum_{i=1}^{N} l\left(y_i, \hat{y}_i\right) + \sum_{k=1}^{K} \left(\gamma T_k + \frac{\lambda}{2} \|w_k\|^2\right)$$

Final Prediction
Predict Final labels $\hat{y}_i = sign\left(f\left(x_i\right)\right)$
Final predicted embryo labels $\hat{y}_i \in \{0, 1\}$

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